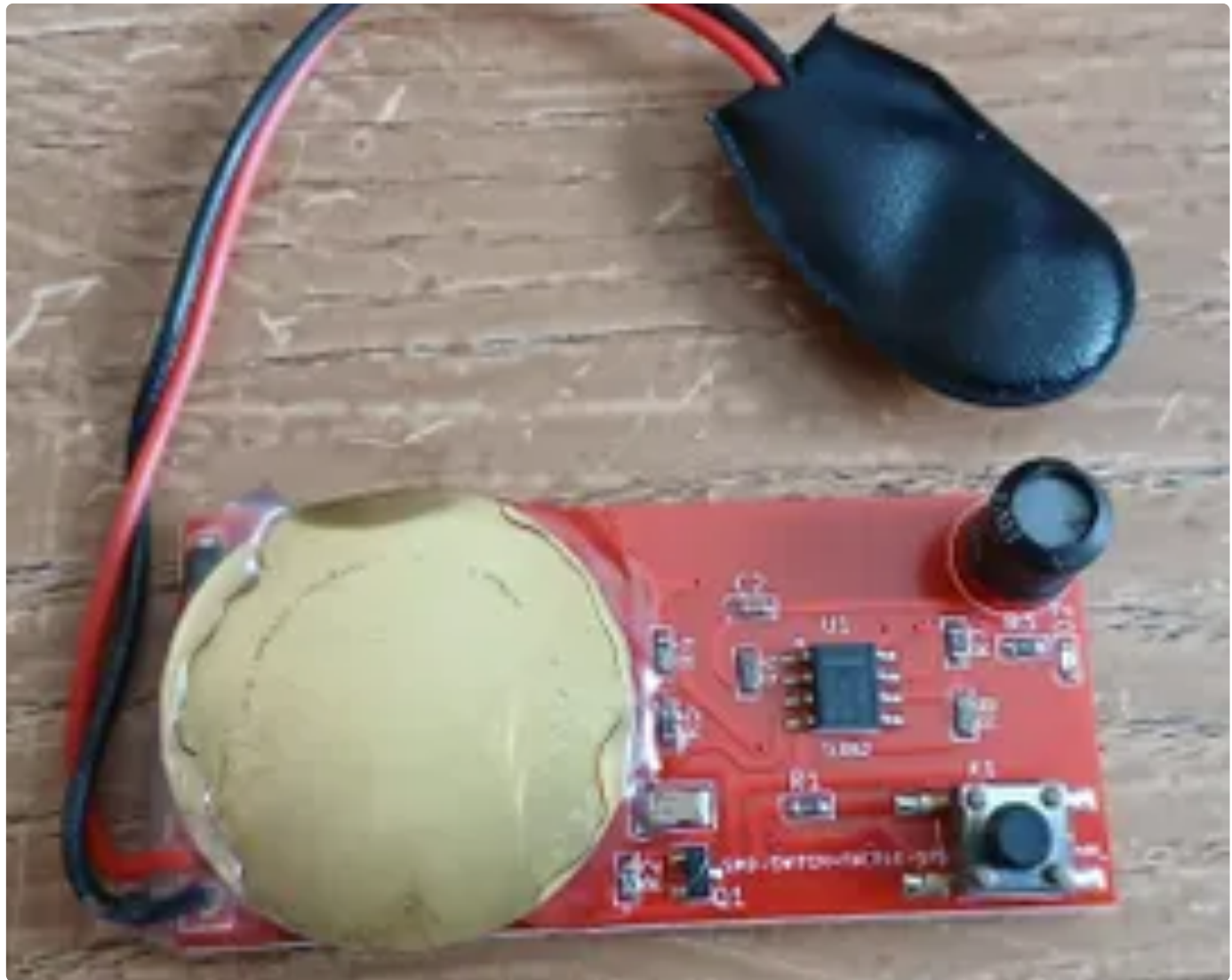


ELF Sensor

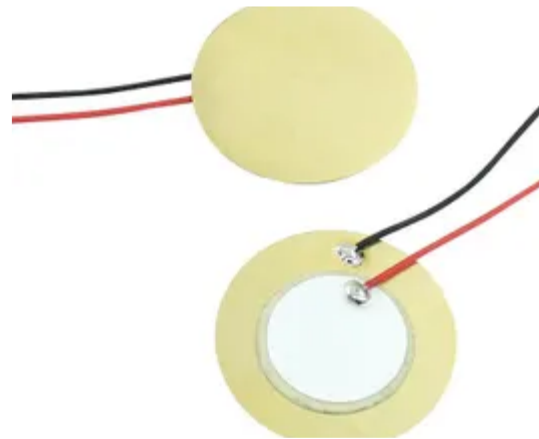
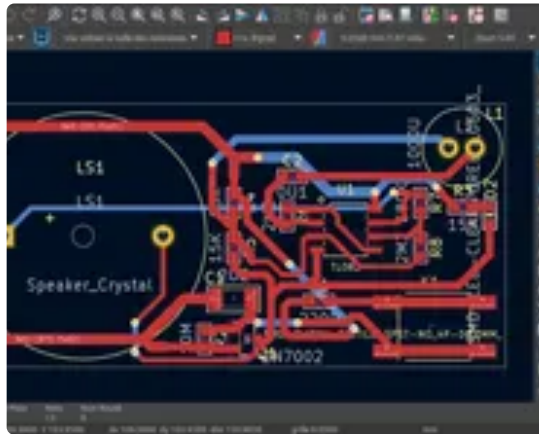
By quicksilv3rflash in Circuits > Tools

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This ELF sensor converts an ambient extremely-low-frequency (ELF) electromagnetic field to an audible tone. Volume scales with field intensity. It does not, unfortunately, detect elves.

Supplies



You will need the [device design files](https://github.com/quicksilv3rflash/ELF_sensor), available at:

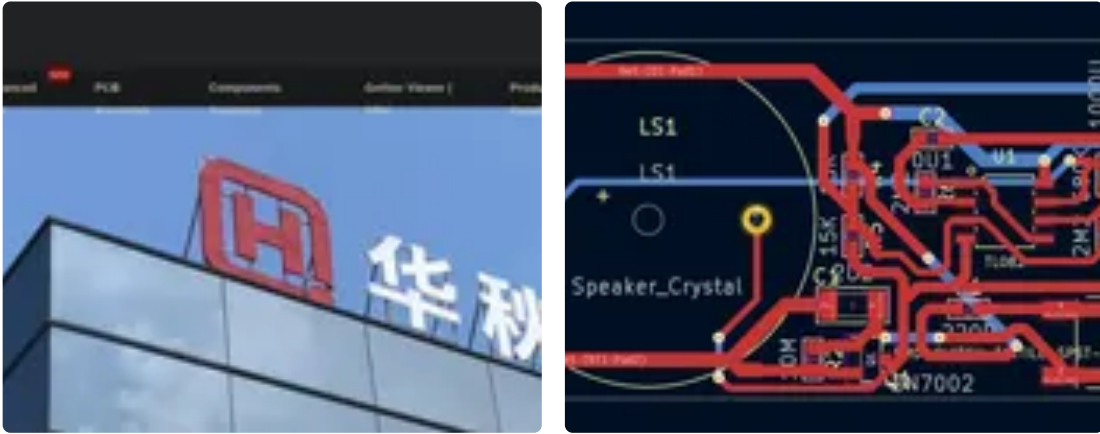
https://github.com/quicksilv3rflash/ELF_sensor

You will also need some [27mm piezoelectric transducers](https://www.amazon.com/Zorfeter-Transducer-Elements-Microphone-Triggers/dp/B0DRFBGDMZ):

<https://www.amazon.com/Zorfeter-Transducer-Elements-Microphone-Triggers/dp/B0DRFBGDMZ>

The buzzer originally specified (LS1), HND-2310B, is not suitable because it contains an internal oscillator circuit. Its datasheet does not make this overly clear, which led to the adventure of desoldering and replacing several hundred of them.

Step 1: Place PCBA Order



Place an order for the design from a PCB / PCBA fabricator such as NextPCB:

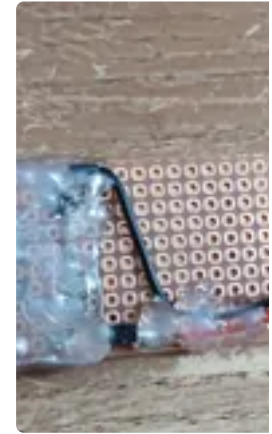
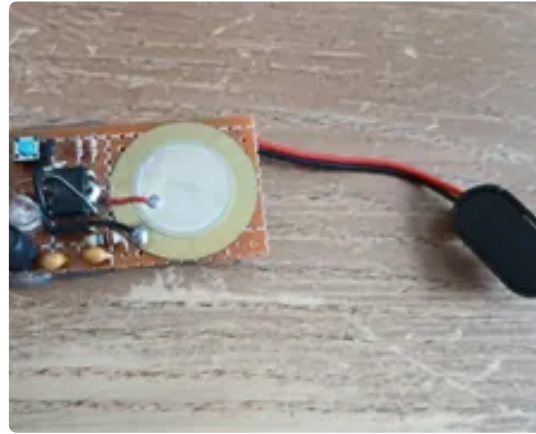
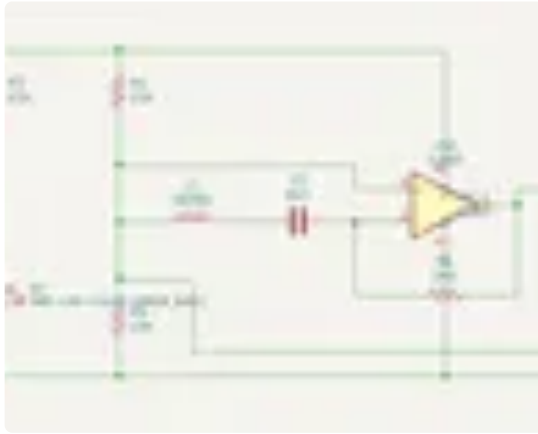
<https://www.nextpcb.com/pcb-assembly-quote#/pcb-assembly-quote>

Step 2: Attach Piezoelectric Transducers



The included bill of materials deliberately leaves out the piezoelectric transducers (because the originally specified one is incorrect for this application), so the boards will arrive with the transducers unpopulated. You will need to solder piezoelectric transducers to the board (or adjust the bill of materials to include a piezo element that does not contain any active circuitry).

Step 3: Design Notes



When the button is pressed, 9V is applied across the gate of Q1, turning it on. This powers the amplifier circuitry. An electromagnetic field, if present, induces a voltage in L1, which is amplified by U1A (an inverting amplifier with a virtual ground), then buffered by U1B, and then applied to the piezoelectric transducer, producing an audible tone at the same frequency as the ambient electromagnetic field. C1 tends to keep Q1 on, while R2 causes it to turn off after about 30 seconds; hold down the button to keep the device on.